Prepared for referee review

TECHNICAL SUPPLEMENT TO NEURAL BELLMAN OPERATORS

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This focused supplement gives the complete construction, complete-policy, signed-risk, reference-centered-risk, continuous investment, and decision-resolution proofs used by the article. It adds proofs of the deterministic adaptive information lower bound, exact representation comparisons, verified readout improvement, and the workload-consistent purchase frontier. Intervening historical proofs are preserved as complete sections in the Numerical and Development Appendix.

1. PROOFS OF THE INHERITED NBO RESULTS

These proofs retain the original statements and arguments. Their placement in the supplement changes the reading order, not their hypotheses.

1.1. *Proof of the result 1*

PROOF: Backward comparison and cash invariance give

$$V_t^* \geq f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J_t^\pi \leq f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$. The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (3). A shift $f_t \mapsto f_t + c_t^0$ adds $c_t^0 - \beta_t c_{t+1}^0$ to u_t and subtracts it from v_t. The terminal width is unchanged. *Q.E.D.*

1.2. *Proof of the result 2*

PROOF: The selected node label lies between the node Bellman infimum minus e_t and that infimum plus $D_t k_t + e_t$. The Lipschitz property of the infimum implies that every cone lies above it minus e_t. At a nearest node its cone exceeds the infimum by

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at most $D_t k_t + e_t + 2L_t h_t$. At the attaining original index, the two feasible pairs and (4) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \le y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (6). The terminal residual belongs to $[-e_T, e_T + 2L_g h_T]$; Proposition 1 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities $|z| = \mathrm{ReLU}(z) + \mathrm{ReLU}(-z)$ and $\min(u, v) = u - \mathrm{ReLU}(u - v)$ give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. *Q.E.D.*

1.3. *Proof of the result 3*

PROOF: The triangle inequality bounds $f(x)$ above by each corner expression. Choose the lexicographically first attaining site $i_*(x)$. In every coordinate choose the cell endpoint between x_j and $x_{i_*,j}$. At this corner distances add exactly. An owner of strictly smaller transformed value would improve the objective at x; a tied smaller owner would contradict the original tie rule. Thus this corner has owner $i_*(x)$, proving both identities. Along a coordinate line, forward and backward relaxations respectively minimize over sites on either side, always comparing value–original-index pairs. Iterating over coordinates gives the separable distance transform and its stated work. Rational arithmetic and unchanged repair complete the claim. *Q.E.D.*

1.4. *Proof of the result 46*

PROOF: The state cost has gradient infinity norm at most $8/d$, Hessian norm at most $22/d$, and diagonal second derivatives at most $5/d + 16/d^2$. For the terminal cost the corresponding bounds are $10/d$, $26/d$, and $9/d + 16/d^2$. They follow by differentiating the quadratic terms and the continuously differentiable squared shortage. The bound on the cycle Laplacian is four. The shortage Hessian, where it exists, is $16\mathbf{1}\mathbf{1}^\top/d^2$; the same second-difference bound holds across its kink.

For feasible (x, a) and (y, b), put $w = x - y$. The difference between the transition at their convex combination and the corresponding combination of transitions is $\theta(1-\theta) w_j w_{j+1}/16$ in coordinate j. Convexity and the G_{t+1} Lipschitz bound

therefore give a convexity defect of at most

$$\frac{\beta G_{t+1}}{16}\theta(1-\theta)\sum_j |w_j w_{j+1}| \le \frac{\beta G_{t+1}}{16}\theta(1-\theta)\|w\|_2^2.$$

The state cost supplies curvature $4/d$, while the two-control action cost has Hessian eigenvalues at least $7/4$. Thus the state curvature of Q_t^* is at least $4/d - \beta G_{t+1}/8$, which is at least $107/(128d)$ when $G_{t+1} \le 27/d$. Partial minimization over the affine capacity graph preserves the state convexity inequality. The minimizing action is unique, including at a binding capacity constraint.

To propagate Lipschitz continuity, write $a = c(x)u$, where u lies in the fixed unit simplex. Transfer the same u from x to y. Each action-cost partial derivative has absolute value at most $13/16$, and $\sum_j u_j \le 1$. The action-cost transfer contributes at most $13\|x-y\|_1/(128d)$. The transition Jacobian at a fixed action has column sums at most $11/16$; the capacity transfer adds at most $3/64$. Comparing a minimizing normalized action in both directions gives the stated recursion for G_t. Its right side maps $27/d$ below itself.

For a fixed u, the transition as a function of x has Jacobian norm at most $11/16 + 1/16 = 3/4$. Its second-order composition term is bounded by $G_{t+1}/8$. The action-cost Hessian has norm at most $21/4$, and the capacity gradient has squared norm $1/(64d)$. These facts give the recursion for M_t. The infimum over u preserves this common semiconcavity bound: after subtracting the same quadratic, each fixed-u function is concave, and an infimum of concave functions is concave. The right side maps $54/d$ below itself. Convexity together with semiconcavity yields the claimed differentiability and gradient regularity; this conclusion does not require a differentiable optimizing-action map.

Along one state coordinate, the fixed-u transition has zero second derivative. The squared norm of its first derivative is at most $85/256 + 23/(256d)$: the two original nonzero entries are bounded by $9/16$ and $1/8$, and the capacity-induced entries are bounded by $1/(16d)$. Combining this with the diagonal stage-cost bound gives D_t. Finally, positive interpolation along one coordinate has error between zero and $D_t h^2/8$. Applying the coordinate interpolations successively and adding the bounds proves (92). *Q.E.D.*

1.5. *Proof of the result 47*

PROOF: Joint convexity, including the feasible graph, gives

$$Q_t^*(x, \bar{\pi}_t(x)) \le \sum_v w_v(x) Q_t^*(v, a_t(v)) \le I_n q_t^{\mathrm{sel},+}(x).$$

The lower interpolation bound gives $V_t^*(x) \ge I_n l_t(x) - \kappa_t$. Subtraction proves the uniform optimal-advantage bound e_t. The partial action derivatives of Q_t^* are bounded by Λ_t, so feasible implementation error adds at most $\Lambda_t b_t$. Adding the future policy loss and proceeding backwards proves the recursion. No convexity of $J^{\widehat{\pi}}$ is used. *Q.E.D.*

1.6. *Proof of the result 36*

PROOF: Apply concavity to $f(a) - Ha^2/2$ and substitute the two lower endpoints. The displayed quadratic has derivative $D - K + 2Ks$. Minimizing it on the unit interval gives the three cases. Its coefficient of K is $-s(1-s)$, which proves the last assertion. *Q.E.D.*

1.7. *Proof of the result 37*

PROOF: At $r = 1$, analytic endpoint enclosures and the minorant lemma give the assertion. Suppose it holds for $r - 1$. Each child interval encloses the true optimal continuation. If the numerical next state is an interval, expanding the queried center value by G_{r-1} times the maximum ℓ^1 displacement encloses all its possible values. Conditional-mean lower bounds and the semiconcavity remainder then give valid endpoint brackets for the true Q_r. The minorants cover the entire action interval. A downward capacity endpoint can omit only an interval of length equal to the certified capacity-rounding difference; subtracting Λ_r times that difference accounts for its possible improvement. The minimum upper endpoint is the cost of a feasible first action followed by the optimal future. This proves (i) by induction. Retaining earlier valid lower bounds preserves every conclusion.

For termination, an interval of length h has minorant at least its smaller lower endpoint minus $H_r h^2/8$. Its smaller upper endpoint is an available global upper bound. With the endpoint-width guard, an interval of length at most $h_\delta = \sqrt{3\delta/H_r}$

cannot be responsible for a gap exceeding δ. Every refined parent is therefore longer than h_δ. Its children are longer than $h_\delta/5$. Apart from the at most two initial fragments made by a prediction, disjoint terminal fragments consequently have total number at most $5c(x)/h_\delta$, up to endpoint counting. Equation (74) is a conservative point budget. Bounded action-quantum rounding of a proposed quarter-to-three-quarter split preserves the one-fifth property whenever the quantum is smaller than $h_\delta/20$. Otherwise precision must be increased; a fixed-precision run is not allowed to ignore this requirement.

For (iii), let $\bar{x}$ be the acquired state. Its downward displacement is at most $d2^{-b}$ in ℓ^1, and $c(\bar{x}) \le c(x)$ preserves feasibility. At a fixed action the stage cost is $8/d$-Lipschitz and the transition has column sums at most $11/16$. Thus replacing $\bar{x}$ by x changes Q_r by at most $(8 + 11\beta d G_{r-1}/16)2^{-b}$. The corresponding value change is at most $dG_r 2^{-b}$. Add these two allowances to (i). Substituting the policy's own future yields the usual one-step loss recursion; backward induction gives (75). No policy-value oracle and no population claim about fitting are used. *Q.E.D.*

1.8. *Proof of the result 27*

PROOF: A routed action is merely another feasible partition point. The endpoint and minorant induction in Theorem 37 therefore applies without change. After each evaluation retain the smallest upper endpoint and the maximum of the old and newly valid global lower endpoints. Their difference cannot increase. The stopping test uses this difference, not a fitted value or a prediction error.

Put $h_\delta = \sqrt{3\delta/H_r}$. A partition interval of length at most h_δ has minimum minorant at least its smaller lower endpoint minus $3\delta/8$. The associated upper endpoint, with the stipulated width and capacity allowance, is within δ of that minorant. Such an interval cannot be responsible for an unresolved global gap. Every split parent therefore has length greater than h_δ. The one-fifth condition makes each of its children longer than $h_\delta/5$. Starting with one interval, disjoint terminal fragments consequently number at most $1 + 5c(x)/h_\delta$. Counting endpoints and allowing for rounding of this numerical bound gives the displayed conservative budget. Floating

failure is handled by the separately charged exact-rational procedure, not by this real-arithmetic count.

The first assertion about predictor evaluation follows from the order of the stopping test and the routing rule. For the last assertion, use backward induction on the remaining horizon, with the common analytic terminal kernel as base. At each nonterminal query, declined predictions leave the same conventional split, lower bound, witness and tie rule. Induction over partition refinements gives the same next query and returned endpoints; the induction over dates then gives the same implemented policy. Model-evaluation work can still differ, so transcript identity is not a zero-cost fitting claim. *Q.E.D.*

1.9. *Proof of Proposition 28*

PROOF: Every retained lower endpoint is at least its true endpoint value minus w, and hence at least $f(a^*) - w$. On an interval of length at most h, the parabolic minorant is therefore at least $f(a^*) - w - Hh^2/8$. Subtracting the omitted-capacity allowance gives $L \geq f(a^*) - w - Hh^2/8 - \Lambda\Delta$; retaining an older, larger valid lower bound only improves this inequality. The upper endpoint at $\widehat{a}$ is at most $f(\widehat{a}) + w$. Semiconcavity and the one-sided first-order condition give $f(\widehat{a}) - f(a^*) \leq \lambda e + He^2/2$. Subtract the lower bound to prove (60). At an interior optimum the derivative is zero. The example $f(a) = a$ on $[0, c]$ shows that the boundary linear term is needed. Finally, retaining bounds makes $U' \leq U$ and $L' \geq L$; expansion proves (61). *Q.E.D.*

1.10. *Proof of the result 24*

PROOF: For each drawn bin sequence, conditional integration over its continuous realizations places the exact path expectation between the interval endpoints. Uniform mixing over the equal bins recovers the original continuous law. The lower endpoint expectation is thus no larger than D_i, and the upper endpoint expectation is no smaller. Apply the bounded empirical-Bernstein inequality of Maurer and Pontil (2009) to the endpoint samples and take the union bound, with outward arithmetic included in the mean, variance and radius. Conditional validity for every

$\mathcal{H}$ implies unconditional validity by integration. On this one simultaneous event, all ensuing deterministic comparisons of the covered quantities hold together. *Q.E.D.*

1.11. *Proof of the result 25*

PROOF: Subtract the gain intervals from k and compare the resulting interval endpoints with zero. Taking the image of $[l, u]$ under the absolute-value map gives (56). The assertions hold simultaneously for every nonnegative fee on the original coverage event. *Q.E.D.*

1.12. *Proof of the result 26*

PROOF: Subtract the common J_{b_0}. The selected implementation has net difference at most $U(\lambda)$, and the best catalogue implementation has net difference at least $L(\lambda)$. This proves the regret bound pointwise on the one simultaneous event and hence for every λ. A minimum of finitely many affine functions changes its active affine expression only at an intersection. For $\lambda_2 > \lambda_1$, sum the two minimizing inequalities to obtain $(\lambda_2 - \lambda_1)(s_{\widehat{b}(\lambda_2)} - s_{\widehat{b}(\lambda_1)}) \le 0$. Since s_b increases strictly with b, the precision comparative static follows. *Q.E.D.*

2. PROOFS FOR COMPLETE-POLICY AND POPULATION TRANSFER

2.1. *Original-law occupation identity*

Fix $\theta \in \Theta_\epsilon$ and use the law generated by its complete feasible policy. Boundedness permits finite summation and conditional expectation. From $B_{t+1} = B_t \beta_t$ and (10),

$$\begin{aligned}\sum_{t<T} B_t \mathbb{E}_{\pi_\theta} A_t^\tau(X_t, \pi_{\theta,t}(X_t)) &= \sum_{t<T} B_t \mathbb{E}_{\pi_\theta} c_t(X_t, \pi_{\theta,t}(X_t)) \\ &\quad + B_T \mathbb{E}_{\pi_\theta} V_T^\tau(X_T) - \mathbb{E} V_0^\tau(X_0) \\ &= J_0^{\pi_\theta} - J_0^\tau.\end{aligned}$$

The conditional-expectation step uses the candidate's transition, not the reference occupation. The terminal equality uses the common original terminal cost $V_T^\tau = g$.

Define y_{tCa} as the indicator that this candidate selects label a in cell C, $m_{tC} = \Pr_{\pi_\theta}(X_t \in C)$, and $q_{tCa} = m_{tC} y_{tCa}$. Define z_{tCaD} as the joint probability of $X_t \in C$,

label a, and $X_{t+1} \in D$. Then the initial, conservation and assignment equations in (13) follow directly. Integrating the pointwise transition bounds against the candidate's subprobability law on C gives the two inequalities for z, including when $q = 0$. An actually realized common parameter cannot realize all the events in a certified forbidden set, so every conflict inequality holds. Consequently this integral point belongs to $\mathcal{P}_\epsilon$. Each stage's advantage expectation is bounded by $\sum_{C,a} \ell_{tCa} q_{tCa}$ and $\sum_{C,a} u_{tCa} q_{tCa}$. Taking extrema over the outer polytope proves Theorem 4.

The polytope is closed and bounded: m_t, q_t are subprobabilities or probabilities, $0 \leq y \leq 1$, and z is bounded by its nonnegative row sum. It is nonempty whenever the qualification class is nonempty, in particular when it contains $\widehat{\theta}$. Hence its extrema exist. Adding valid constraints shrinks it. If a smaller tolerance retains all larger-tolerance constraints and possibly more, its lower extremum increases and its upper extremum decreases. Independent dual feasibility supplies the appropriate weak-duality bound even when the numerical solve is incomplete. Adding and subtracting the two reference costs proves the class comparison. These operations are deterministic on the simultaneous input event.

Why reference weights are insufficient. Consider a two-date, zero-stage-cost example. Initially only state s is visited. At date zero, action L moves to state ℓ and R to state r. At date one the costly action costs one at r and zero at ℓ. The reference selects L; a candidate selects R and then the costly action at r. The reference's date-one occupation gives r mass zero, but the candidate's cost difference is one. A first-date formula using the reference continuation sees zero date-zero contrast and omits the later change. The flow constraint instead forces $m_{1r} = 1$ and charges the date-one advantage. This example identifies an error that cannot be repaired by a union bound over cells.

Shared coefficients remain material. In the same example, let $\theta \in \{-1, 1\}$. At s action R has score $-\theta$ and L score zero; at r the costly action has score θ and the free action score zero. Each event is individually possible, but R and the costly action require opposite signs. Every common-parameter policy has cost zero, whereas the unconstrained action relaxation permits cost one. The exact conflict inequality excludes their simultaneous integral choice. Its fractional relaxation can still admit

cost 1/2; this is why the theorem asserts validity and monotone tightening, not exact attainability of every relaxed occupation.

2.2. *Nonlinear evaluations and strategic deviations*

Assume inductively that $V^{\tau}_{t+1} + e^{-}_{t+1} \le V^{\pi}_{t+1} \le V^{\tau}_{t+1} + e^{+}_{t+1}$. Monotonicity and cash invariance give

$$\mathcal{T}^{\pi}_{t} V^{\tau}_{t+1} + \beta_t e^{-}_{t+1} \le V^{\pi}_{t} \le \mathcal{T}^{\pi}_{t} V^{\tau}_{t+1} + \beta_t e^{+}_{t+1}.$$

Subtract V^{τ}_{t} and add the stated residual bounds. The common terminal condition starts the induction. Under a general continuation Lipschitz modulus instead, triangle inequality gives $\|V^{\pi}_{t} - V^{\tau}_{t}\|_{\infty} \le b_t + \lambda_t \|V^{\pi}_{t+1} - V^{\tau}_{t+1}\|_{\infty}$; iteration proves the second formula of Proposition 5 on the stipulated invariant domains.

For player i, fix every other player's complete strategy. The induced controlled kernel and the player's feasible actions define a best-response problem, with the player's own stated recursive objective. Its best-response value is below the cost of the attained action sequence. Apply the one-sided residual sandwich to this problem from every date and state. A bound r_{it} on its difference is exactly a bound on that player's profitable unilateral cost reduction. Applying it simultaneously to every player and subgame establishes the claimed approximate deviation property. This argument does not establish existence of a profile satisfying the premises. For temporal selves, each self is a separate player with its own deviation problem; no identity with a commitment planner's optimum is asserted.

If a numerical kernel $\widetilde{\mathcal{T}}^{\pi}_{t}$ differs uniformly from the original kernel by at most κ_t on the continuations in use, it adds κ_t to the stage residual. Its contribution is therefore at most $\sum_t (\prod_{j<t} \lambda_j)\kappa_t$. To compare a numerical policy with the original optimum, the corresponding optimal-operator approximation must also be charged in the two-sided sandwich. This explicitly accounts for a diffusion approximation without postulating an unproved rate in its time step.

3. PROOFS FOR JOINT-DOMAIN CONSTRUCTION AND EXPLORATION

3.1. *One population circuit*

Let $G_t(r) = \mathcal{T}_t f_{t+1}(r)$ on the joint domain. Primitive regularity and the Hausdorff action bound imply that it is L_t-Lipschitz by the same comparison as in Theorem 2, now including the context coordinates. For each node, $-e_t \le y_{ti} - G_t(r_{ti}) \le D_t k_t + e_t$. Thus every cone is at least $G_t(r) - e_t$, and a node within h_t gives a cone at most $G_t(r) + D_t k_t + e_t + 2L_t h_t$. The lower envelope has Lipschitz modulus L_t because each of its members does. The original label owner, the feasible repair, and the original action modulus give the selected-policy upper residual $e_t + D_t \zeta_t$ exactly as in the original construction. The argument closes backward from the terminal approximation, and then the original policy sandwich proves the loss account.

Use $|u| = \mathrm{ReLU}(u) + \mathrm{ReLU}(-u)$ and $\min(u, v) = u - \mathrm{ReLU}(u - v)$. Compute all cone values and combine them in a balanced minimum tree. This proves the claimed weight/operation count and depth, up to the fixed affine layers. Lexicographic original-owner selection is a separate measurable comparison operation; the discontinuous actor is not falsely called a continuous ReLU function. Its feasible repair is charged separately. The guarantee holds at every point of every joint domain, so evaluating it along an endogenous sequence uses no distributional inference or independence assumption. The covering exponent and effective construction cost remain explicit.

3.2. *Adaptive acquisition coverage*

Fix one cover ball B. Let E_i be the event that the first i acquisitions miss it. By the conditional lower-probability assumption,

$$\Pr(E_i) = \mathbb{E}[\mathbf{1}_{E_{i-1}} \Pr(Y_i \notin B \mid \mathcal{F}_{i-1})] \le (1 - a_h) \Pr(E_{i-1}).$$

Induction yields $(1 - a_h)^n$. A union bound over the M fixed balls gives the result. If none is missed, any point is within h of a cover center and that center is within h of an acquired node, hence the radius is $2h$. This argument controls geometry, not label error; any probabilistic label-certification event is included separately in the

overall error allocation. Pure exploitation without a conditional lower-probability bound does not satisfy the proposition.

3.3. *Prediction and fallback work*

Suppose the best action is a_* and $\Delta > 4e$. For every competitor b, its estimate exceeds that of a_* by at least $\Delta - 2e > 2e$. The estimate-minimizing action is consequently a_*, and its upper confidence endpoint is below every competitor's lower endpoint. Thus failure of the screen implies $\Delta \le 4e$. Its probability is at most the tie mass plus $C(4e)^\alpha$, truncated at one. Multiplying by the fallback's upper work bound and adding proposal and test overhead proves Proposition 3. To certify a saving one must compare that upper account with a lower account for the conventional baseline, not with another upper bound. This distinction is the work analogue of not subtracting policy-loss upper bounds to estimate a cost difference.

4. PROOF OF THE DATED ACCURACY ALLOCATION

Hölder's inequality gives, for every feasible positive allocation,

$$\begin{aligned} S &= \sum_t (a_t \eta_t^{-p})^{1/(p+1)} (w_t \eta_t)^{p/(p+1)} \\ &\le \left(\sum_t a_t \eta_t^{-p} \right)^{1/(p+1)} \left(\sum_t w_t \eta_t \right)^{p/(p+1)}. \end{aligned}$$

Therefore the objective is at least $S^{p+1} \varepsilon^{-p}$. Substituting (97) attains equality and uses the whole error budget. Strict convexity of $\eta \mapsto \eta^{-p}$ gives uniqueness. The discounted expression follows by summing the geometric series with ratio $\beta^{p/(p+1)}$; the undiscounted expression follows from $S \le (T+1) A^{1/(p+1)}$. For the clipped allocation $\eta'_t = \min(\eta_t^*, \eta_{\max})$, the error budget can only decrease and

$$a_t (\eta'_t)^{-p} = a_t \max\{(\eta_t^*)^{-p}, \eta_{\max}^{-p}\} \le a_t (\eta_t^*)^{-p} + a_t \eta_{\max}^{-p}.$$

The asserted additive bound follows. Dyadic downward rounding by less than a factor two proves the final factor. Bit complexity, per-date setup and storage costs lie outside this algebraic upper model unless expressly included in a_t.

5. PROSPECTIVE MENU CHOICE AND UNIFORM ECONOMIC SENSITIVITY

5.1. *Finite-menu selection*

Condition on everything used to construct the finite service menu. Let $C_j(W) \in [0, H]$ be the true cost of service j on a new IID workload W, and suppose $l_{ji} \leq C_j(W_i) \leq u_{ji}$ for each evaluated workload. The endpoints are valid whole-input enclosures, not unqualified point calculations. A bounded-variable exponential inequality and a union bound over the m services give, with probability at least $1 - \delta$,

$$|\overline{C}_j - J_j| \leq q_n, \quad q_n = H\sqrt{\log(2m/\delta)/(2n)}, \quad j = 1, \ldots, m.$$

For completeness, the conditional moment-generating-function bound for a centered variable in an interval of length H is at most $\exp(\lambda^2 H^2/8)$. Iterating it over the independent draws, applying Markov's inequality and optimizing λ proves each tail $\exp(-2nq_n^2/H^2)$. Hence $J_j \in [\bar{l}_j - q_n, \bar{u}_j + q_n]$ simultaneously.

Let service $\widehat{j}$ minimize $\lambda\bar{u}_j + p \cdot w_j + f_j$ for $\lambda \geq 0$, and let j_* minimize its true counterpart. On this event, the selected objective minus the true optimum is at most

$$\begin{aligned} &\lambda(\bar{u}_{\widehat{j}} + q_n) + p \cdot w_{\widehat{j}} + f_{\widehat{j}} - \lambda(\bar{l}_{j_*} - q_n) - p \cdot w_{j_*} - f_{j_*} \\ &\quad \leq \lambda\{2q_n + \bar{u}_{j_*} - \bar{l}_{j_*}\} \leq \lambda\{2q_n + \max_j(\bar{u}_j - \bar{l}_j)\}. \end{aligned}$$

The same event works for every price and fee vector simultaneously, because it was constructed for the costs before prices were applied. This does not make an outcome-dependent acquisition or construction of new services free of a selection charge. On a separate evaluation stream, simultaneous pairwise intervals give the sharper evaluated regret bound $\max\{0, \max_j u_{\widehat{j},j}\}$ for the calibration-selected service. The stream need not show strict superiority to establish noninferiority at a prespecified margin.

For adaptively generated workloads, iteration of the conditional moment-generating-function bound instead bounds the average predictable cost $n^{-1}\sum_i \mathbb{E}[C_j(W_i) \mid \mathcal{F}_{i-1}]$, for a menu fixed before that sequence. It does not identify a stationary population expectation or the cost under another policy's counterfactual

workload distribution. This is the correct distinction between sequential validation and an IID new-task claim.

5.2. *Price and fee regions*

For a compact joint cost polytope $\mathcal{D}$ and two feasible assignments x, y, the largest objective difference is

$$\sup_{c\in\mathcal{D}}\{\lambda c\cdot(x-y)+p\cdot(w_x-w_y)+f\cdot(x-y)\} = \lambda h_{\mathcal{D}}(x-y)+p\cdot(w_x-w_y)+f\cdot(x-y).$$

Nonnegativity of λ is used here. The candidate x is optimal for every $c\in\mathcal{D}$ precisely when this expression is nonpositive for every feasible y. This proves the polyhedral region in Proposition 7; a certified support upper bound gives a sufficient subregion. Linear optimization over a compact price-fee polytope reduces uniform verification to its vertices. The boundaries are the stated price-fee breakpoints. If $\mathcal{D}$ includes correlated pairwise constraints they must be retained in its support problem; independently choosing the worst endpoint of each pair can be looser. No step estimates a price, monetary scale or adoption fee from a clock measurement.

6. SIGNED QUALIFICATION AND NONLINEAR TRANSPORT: PROOFS

6.1. *Proof of Theorem 6*

Fix θ and omit it from the notation. Because b_t^- is valid for every original feasible action, taking infima gives

$$T_t f_{t+1}-f_t \geq \widehat{T}_t-f_t+b_t^- \geq \ell_t+b_t^-.$$

For the actual feasible action, the numerical greediness and actor-specific upper bounds give

$$T_t^{\pi} f_{t+1}-f_t \leq \widehat{Q}_t(\cdot,\pi_t)-f_t+b_t^{\pi,+} \leq u_t+\gamma_t+b_t^{\pi,+}.$$

Define real constants backwards by

$$a_T=d^-,\quad c_T=d^+,\qquad a_t=\ell_t+b_t^-+\beta_t a_{t+1},\quad c_t=u_t+\gamma_t+b_t^{\pi,+}+\beta_t c_{t+1}.$$

At the terminal date $f_T + a_T \leq g \leq f_T + c_T$. If $V^*_{t+1} \geq f_{t+1} + a_{t+1}$, monotonicity and cash invariance, including under the infimum, imply

$$V^*_t = T_t V^*_{t+1} \geq T_t f_{t+1} + \beta_t a_{t+1} \geq f_t + a_t.$$

Similarly, $V^\pi_{t+1} \leq f_{t+1} + c_{t+1}$ implies

$$V^\pi_t = T^\pi_t V^\pi_{t+1} \leq T^\pi_t f_{t+1} + \beta_t c_{t+1} \leq f_t + c_t.$$

Induction proves both comparisons. Subtracting at date zero gives $V^\pi_0 - V^*_0 \leq c_0 - a_0$, which expands to (16). Feasibility and optimality give its lower bound zero. Infima need not be attained: the order comparison holds directly under infima, provided the stipulated original value recursion and feasible policy are defined. Every parameter is treated by the same argument; a supremum of valid bounds provides the stated classwise conclusion. No union over parameter vectors and no reference-policy occupation law is introduced.

For shifts $f'_t = f_t + k_t$, set $\widehat{Q}'_t = \widehat{Q}_t + \beta_t k_{t+1}$. Both numerical residual endpoints shift by $\beta_t k_{t+1} - k_t$; their difference is unchanged. Numerical action differences and original-minus-numerical action values are unchanged by cash invariance. Both terminal endpoints shift by $-k_T$. Thus every term in (16) is invariant. If the action-value discrepancy itself is a constant k_t, choose $b^-_t = b^{\pi,+}_t = k_t$ and its charge vanishes.

For the stationary assertion, the Bellman and fixed-policy maps are sup-norm contractions with modulus β: monotonicity and cash invariance imply sup-norm nonexpansiveness of each one-step evaluation. The finite-horizon argument with a bounded terminal comparison then has a terminal remainder of order β^T. Letting T tend to infinity proves the displayed stationary bound. The asserted constant-shift invariance does not justify shifting an original preference primitive or an action-dependent immediate cost.

6.2. *Sharpness and the distinction between constant and statewise bias*

For $h > 0$, take one decision date, terminal cost $g(y) = y$, discount one and zero immediate costs. The original successors of actions zero and one are respectively

h and zero; the numerical successors are both zero. Store $f_0 = 0$ and $f_1 = g$, and let the numerical tie convention select action zero. All numerical residual and greediness terms vanish. The original-minus-numerical action values lie in $[0, h]$, the selected-action upper endpoint is h, and the implemented policy loses exactly h. Thus the signed charge h is attained. The symmetric allowance $2h$ remains valid but is strictly larger. The example works for any monotone cash-invariant normalized risk evaluation because its successors are deterministic. Both laws can be singular, so no total-variation convergence is being used.

A statewise common-action bias is different. Let a root action choose state L or R at the next decision date. At that date there is only one feasible action, with original cost $2h$ in L and zero in R. Let every numerical cost, stored continuation and terminal cost be zero, with discount one. Break the root numerical tie toward L. All numerical residual spans and greediness gaps vanish. At the second date the discrepancy is common to every action at its state, but equals $2h$ or zero depending on that state. The implemented policy loses $2h$. Cancelling the discrepancy merely because it is action independent at each state would falsely report zero loss. The signed uniform range correctly charges $2h$. This is precisely the earlier action's effect on future states; fixing the reference occupation would also lose it.

6.3. *Proof of Proposition 7*

Let $(Y, \widehat{Y})$ have any coupling of the two successor laws. For every real threshold z, the map $r \mapsto (r - z)_+$ is one-Lipschitz. Hence

$$\left| P(f - z)_+ - \widehat{P}(f - z)_+ \right| \leq \mathbb{E}|f(Y) - f(\widehat{Y})| \leq K\mathbb{E}d(Y, \widehat{Y}).$$

The same bound, divided by $1 - \alpha$, is uniform in z for the two functions minimized in (17). Infima of uniformly close functions are close. Taking the infimum over couplings proves the first bound, with approximating couplings sufficient when an optimum is not attained. At $\alpha = 0$ the variational expression equals expectation: the expression is at least Pf and equality holds for any threshold no larger than the bounded loss's infimum.

To verify that the tail factor cannot be uniformly reduced, take $f(y) = y$ on $[0,1]$, $\widehat{P} = \delta_0$ and $P = (1-h)\delta_0 + h\delta_1$, with $0 < h \leq 1$ and $\alpha = 1 - h$. Then $K = 1$, $W_1(P, \widehat{P}) = h$, $\rho_P^\alpha(f) = 1$ and $\rho_{\widehat{P}}^\alpha(f) = 0$. The bound $Ke/(1-\alpha) = 1$ is attained. In particular, a bound uniform up to risk level one cannot be deduced from Wasserstein distance alone with a fixed coefficient.

For the entropic assertion, choose $m \leq f \leq M$. For positive λ, the mean value theorem gives

$$|\exp(\lambda f(y)) - \exp(\lambda f(\widehat{y}))| \leq \lambda \exp(\lambda M) K d(y, \widehat{y}).$$

Both exponential integrals are at least $\exp(\lambda m)$, and $|\log s - \log t| \leq |s - t| / \min(s,t)$ for positive s, t. Combining these facts, dividing by λ and taking the infimum over couplings yields $K \exp\{\lambda(M-m)\} e$. Letting m and M approach the continuation's infimum and supremum proves (18). Translation invariance follows from the oscillation rather than the absolute level. The expectation limit follows from boundedness and expansion of the exponential integral; its displayed coefficient tends to K.

If $f(Y) \geq f(\widehat{Y})$ almost surely, every positive-part integral is weakly larger and so is its minimized variational expression. Exponentiation and logarithm preserve the same order for the entropic evaluation. This gives the signed lower endpoint zero. If $|\widehat{Q} - c - \beta \rho_{\widehat{P}}(f)| \leq \xi$, subtracting the numerical approximation places the action-value discrepancy in $[-\xi, \beta\kappa e + \xi]$ under that ordering, or in $[-\beta\kappa e - \xi, \beta\kappa e + \xi]$ without it. These are bounds for the original risk primitive; no statement is made about replacing one primitive by another.

6.4. *Nested objectives and the dated resource account*

Nested one-step evaluations and a static terminal evaluation differ even without decisions. With four equiprobable terminal losses $0, 0, 0, 4$, group the first two and last two under two equiprobable intermediate states. At risk level one half, the intermediate evaluations are zero and four, and the nested value is four. Static average value at risk of the four terminal outcomes is two. Consequently a static

objective needs its own state augmentation or a different analysis; applying the preceding nested recursion without that step changes the objective.

For an originally nested criterion, the preceding induction propagates a state-independent continuation allowance a as $\beta_t a$. The local Wasserstein-to-risk coefficient κ_t enters only the current discrepancy. Substituting the symmetric signed interval into (16) gives (19); monotone-coupling information replaces the factor two on that transition component by one. This establishes the budget used for dated allocation. It does not bound the cost of obtaining K_{t+1} or the number of labels, innovations or feasible-action searches. In particular, a growth of the verified continuation modulus with horizon is retained in κ_t and cannot be removed by this algebra.

For a fixed-opponent best-response problem the identical induction compares a player's implemented complete strategy with the original best-response optimum. Applying it separately in every declared subgame gives the corresponding unilateral-deviation certificate. Each temporal self likewise retains its own original criterion and fixed future strategies. Monotone but non-cash-invariant aggregators do not satisfy the constant-shift step above and must instead use their certified absolute-error recursion. This establishes the advertised scope without conflating expectation, nested risk, static risk and strategic evaluation.

7. PROOFS FOR REFERENCE-CENTERED RISK QUALIFICATION

7.1. *Centering and the original Bellman optimum*

For any bounded z, substituting $f_{t+1} + z$ in (20) and subtracting the finite number $\rho(f_{t+1})$ proves (22). Each dual objective at f_{t+1} is at most its supremum, giving $k^f \geq 0$. Approximating that supremum gives $\inf_q k^f(q) = 0$ without an attainment assumption. Since each q is a probability, adding a constant b to z adds b to $\mathcal{E}^f(z)$. If $z \leq w$, every dual objective at z is no larger than its counterpart at w, proving monotonicity. The inequalities $w - \|z - w\|_\infty \leq z \leq w + \|z - w\|_\infty$ prove nonexpansiveness.

PROOF OF THEOREM 8: Let $d_t^\pi = V_t^\pi - f_t$ and $d_t^* = V_t^* - f_t$. The original recursions and (22) give the exact identities

$$d_t^\pi(x) = r_t(x, \pi_t(x)) + \beta_t \mathcal{E}^f_{t,x,\pi_t(x)}(d_{t+1}^\pi),$$
$$d_t^*(x) = \inf_{a \in A_t(x)} \{r_t(x,a) + \beta_t \mathcal{E}^f_{t,x,a}(d_{t+1}^*)\}. \tag{1}$$

Both have terminal value $g - f_T$. Suppose $d_{t+1}^\pi \le U_{t+1}$. Monotonicity, nonnegative β_t, the upper residual inequality and (23) imply $d_t^\pi \le U_t$. Similarly $L_{t+1} \le d_{t+1}^*$ implies $L_t \le d_t^*$. Backward induction establishes both statements. Feasibility of the supplied complete policy implies $V_t^* \le V_t^\pi$, giving (24). The argument takes place at each fixed parameter vector; applying it to every vector satisfying the premises proves the class statement without assembling incompatible continuations.

Exact equality recursions coincide with (1), proving the exactness assertion by induction. For the scalar comparison, set $a_t = \ell_t + b_t^-$ and $b_t = u_t + \gamma_t + b_t^{\pi,+}$ from the signed assumptions. Every original action has residual at least a_t, since its numerical action value is at least the numerical Bellman infimum. The implemented action has residual at most b_t. Define $\underline{h}_T = d^-$, $\overline{h}_T = d^+$, and

$$\underline{h}_t = a_t + \beta_t \underline{h}_{t+1}, \qquad \overline{h}_t = b_t + \beta_t \overline{h}_{t+1}.$$

Since $\mathcal{E}^f(b) = b$ for constants, these are feasible lower and upper certificates. Exact backward evaluation is order preserving, so $L_t \ge \underline{h}_t$ and $U_t \le \overline{h}_t$ for exact residuals and terminal discrepancies. Their date-zero difference is at most (16).

For two policy supersolutions, their pointwise minimum remains a supersolution: the operator at the minimum of their successor bounds is at most its value at either bound. The corresponding pointwise maximum of two optimum subsolutions remains a subsolution by the reverse argument. Finite combinations therefore preserve an already valid scalar guarantee. No claim about improved bounds from an uncontrolled numerical optimizer is needed. *Q.E.D.*

7.2. *Risk occupations and certificate orientation*

PROOF OF PROPOSITION 9: For fixed π, (1) is the Bellman equation for a finite-horizon maximizer selecting a risk index j after observing the current state. Its

reward is $r_{ts} - \beta_t k_{tsj}$, transition vector is q_{tsj}, and terminal reward is $g - f_T$. Finite maxima have measurable maximizers. Backward induction establishes a Markov maximizing choice at each date and state.

Any nonanticipating choice rule, including a history-dependent one, induces occupation masses satisfying (25). Conversely, for a feasible flow, choose j at (t,s) with probabilities y_{tsj}/m_{ts} when $m_{ts} > 0$; at zero-mass states choose an arbitrary feasible index. Induction on t recovers the declared masses. The corresponding expected sum of rewards is exactly the objective in (26). Thus the flow maximum equals the centered fixed-policy value. This representation is under the adversarial risk kernels and must not be called the physical occupation law.

Multiply each dual inequality by $B_t y_{tsj}$ and sum. The flow equalities cancel intermediate potential terms, leaving the primal objective at most $\mu^\mathsf{T} U_0$. The argument allows a zero discount: it uses $B_{t+1} = B_t \beta_t$ but never divides by either factor. Feasible primal and dual witnesses with equal objectives certify the maximum. For an optimum subsolution, a separate feasible risk witness for every original action supplies a lower bound on that action's supremum; a minimization over a restricted sampled action set does not supply the required all-action inequality. *Q.E.D.*

7.3. *Kernel exclusion and its sharp boundary*

For any $z \in \mathcal{D}$, the difference between the affine pieces indexed by q and q_0 is

$$(q - q_0)^\mathsf{T} z - k^f(q) + k^f(q_0).$$

A strict negative upper bound excludes q uniformly. This proves (27). Maximizing coordinate by coordinate gives the box formula. For a span ball, the coefficients sum to zero; adding a constant to z changes nothing. Normalize $\min_i z_i = 0$ and assign s to the positive coefficients and zero to the negative ones. The maximum is $s \sum_i (q_i - q_{0i})_+ = s\|q - q_0\|_1/2$.

As a sharp example, take two equally likely outcomes and average value at risk at level $1/2$. Then $\rho(v) = \max(v_1, v_2)$ and the dual vertices are e_1, e_2. At $f = (0,1)$ their centered penalties are $1, 0$. For $z = (2,0)$, the centered value is 1. Keeping only the maximizing kernel e_2 from the reference gives 0, which is invalid as an upper

bound; dropping penalties gives 2, which is conservative but not exact. On the span ball of radius $1/2$, the first kernel is safely excluded. At radius 1, $z = (1, 0)$ makes the two pieces equal, so the strict exclusion must retain the boundary. This example proves both the necessity of allowing the risk kernel to change and sharpness of the strict threshold convention.

7.4. *Entropic duality without changing the objective*

Let p be a probability measure, f, z bounded, and $\lambda > 0$. Boundedness makes $Z_f = \int e^{\lambda f} dp$ strictly positive and finite, and makes p_f equivalent to p on its support. For $q \ll p$ with finite relative entropy,

$$\lambda^{-1}\mathrm{KL}(q\|p_f) = \lambda^{-1}\mathrm{KL}(q\|p) - \int f\, dq + \lambda^{-1}\log Z_f.$$

If q places mass outside the support of p, its penalty is infinite. For any bounded v, define $p_v = e^{\lambda v} p / Z_v$. Nonnegativity of $\mathrm{KL}(q\|p_v)$ gives

$$\int v\, dq - \lambda^{-1}\mathrm{KL}(q\|p) \le \lambda^{-1}\log Z_v,$$

with equality at $q = p_v$. This proves the variational representation, its centered penalty, and (28). It also shows exactly why using the fixed tilt p_f as an expectation is not generally exact: the centered evaluation remains entropic under p_f, rather than linear. For $\lambda \downarrow 0$, bounded convergence gives the expectation limit. Exponential and logarithmic evaluations in a numerical certificate require outward arithmetic; the rational average-value-at-risk experiments are not evidence of exact floating-point entropic computation.

7.5. *Occupation-sensitive precision and a strict improvement*

For fixed penalties and transitions, write the flow objective as $A(y)$ and the additional nonnegative residual contribution as $E(y) = \sum_{t,s} B_t e_{ts} m_{ts}$. If $H = \max_y A(y)$ and $H_e = \max_y \{A(y) + E(y)\}$, then

$$0 \le H_e - H \le \max_y E(y).$$

The upper inequality follows from $A(y) \leq H$ for every feasible y; the lower follows from $E(y) \geq 0$. Since each occupation layer has total mass one, $\max_y E(y) \leq \sum_t B_t \max_s e_{ts}$. This establishes (29). The argument assumes the penalties are held fixed while residual uncertainty is enlarged. Altering the underlying stored continuation changes the penalties and requires requalification.

For a strict horizon example, let two states be absorbing under both actions, let action zero cost zero and action one cost one in both states, let $g = 0$, and let all discounts equal one. Every normalized monetary risk evaluation of these deterministic transitions agrees with evaluation at the absorbing state. Put $f_t(s) = (T - t)s$ for $s \in \{0, 1\}$ and choose action zero everywhere. Each date's implemented and minimum residuals are $(0, -1)$; the signed scalar account charges one at each date, yielding T. The exact state-dependent lower and upper recursions both equal $-(T - t)s$, giving zero policy loss at every state. There are two feasible actions and every action is checked. No state-dependent bias was cancelled without propagation: its entire state dependence was carried through the original transition at every date. This is consistent with, and sharpens rather than contradicts, the statewise-bias warning in Section 5.

Finally, the finite backward recursion evaluates each stored transition support once for each retained risk piece at each date and state-action pair. Summing these support sizes yields the stated arithmetic count. A direct descendant recursion repeatedly evaluates the same subproblem; removing that repetition does not establish a polynomial bound in the number of continuous states, risk pieces, acquisition cells or rational bit length. The comparison is a representation-conditioned resource statement, and the full construction and support-qualification work remains separately charged.

8. CONTINUOUS INVESTMENT: PRIMITIVE PROOFS

8.1. *Curvature, the action constraint, and state moduli*

PROOF OF PROPOSITION 10: For $x \in X$, $0 \leq a \leq 1/4$ and $|z| \leq 1/32$, the transition remains in X. Indeed, the state part is increasing in both of its arguments, its maximum is $1/16 + 1/2 + 1/8$, and $b_j a \leq 1/8$. Its minimum after the shock is at

least $1/32$, and its maximum is at most $27/32$. Thus every derivative bound below is used on its stated domain.

Both S and g are convex and continuously differentiable; their squared hinges have positive semidefinite second derivative on each smooth region. Direct differentiation gives

$$|\partial_j g| \le \frac{10}{d}, \qquad |\partial_j S| \le \frac{15}{2d}, \qquad \nabla^2 S \succeq \frac{4}{d} I.$$

The first bound adds $5/d$ from the target term, $1/d$ from the cyclic difference term, and $4/d$ from the hinge. Only $-x_j x_{j+1}/16$ in F_j has nonzero second derivative. For a joint direction (v, w), the possibly negative part of the second derivative of $g(F)$ is

$$-\frac{1}{8}\sum_j \partial_j g(F) v_j v_{j+1} \;\ge\; -\frac{5}{4d}\|v\|_2^2,$$

since $\sum_j |v_j v_{j+1}| \le \|v\|_2^2$. The remaining term is the quadratic form of the positive semidefinite Hessian of g applied to the full state-action Jacobian. In particular, it includes the mixed state-action terms and is not discarded selectively. Since $h''(a) = 2 + 48a^2 \ge 2$, every fixed-shock objective satisfies (32). Integrating against a fixed feasible risk density preserves this bound. Taking the supremum over those densities preserves the same strong convexity. The bound extends across the hinge by continuity of the first derivative, or by integrating the almost-everywhere second-derivative inequality along a segment.

The graph $\{(x, a) : x \in X, 0 \le a \le A(x)\}$ is convex because A is affine. Partial minimization of the jointly convex objective over this graph proves convexity of V_1. Compactness and continuity supply an attained minimum. The positive capacity permits constraint qualification in the action variable.

For a fixed feasible density, the supporting strong-convexity inequality gives, for every (y, u),

$$Q_1(y, u) \ge W + p_x^\top (y - x_0) + p_a(u - a_0) + (u - a_0)^2.$$

For feasible u, adding $\lambda(u - A(y)) - \nu u \le 0$ to the right side preserves a lower bound. Minimize this augmented quadratic over all real u, an enlargement of the feasible interval. Completing the square gives (34). This proof uses neither equality

in the risk supremum nor exact stationarity. Interval implementations bound the complete expression outwards, including its residual square and its slope.

For completeness we bound true value subgradients, rather than the potentially large slopes of arbitrary poor lower witnesses. A transition Jacobian column has nonnegative entries summing to at most $11/16$. Therefore any state subgradient of a risk-maximizing one-date objective has coordinate magnitude at most

$$\frac{1}{d}\left(\frac{15}{2}+\beta\frac{55}{8}\right).$$

At an upper-bound action, the multiplier can be chosen with $0 \le \lambda \le \beta(10/d)\sum_j b_j = 15\beta/4$: the nonnegative derivative of h cannot increase the negative derivative which this multiplier offsets. At a lower-bound action take $\lambda = 0$. At an interior minimizer both action-bound multipliers vanish. Envelope subgradients thus have coordinates bounded by

$$\frac{1}{d}\left\{\frac{15}{2}+\beta\left(\frac{55}{8}+\frac{15}{32}\right)\right\} < \frac{15}{d}.$$

Where the risk function is nonsmooth, the same bound holds for convex combinations of active risk gradients and valid KKT multipliers. Integrating bounded subgradients along a segment, with the one-sided limit at the boundary of X, proves (33). *Q.E.D.*

8.2. *Exact one-date risk and outward construction*

Put $m = F(x,a,0)$, $R = 1/32$ and $D = 10d^{-1}\sum_j m_j s_j$. Since $\sum_j s_j = 0$, the hinge in $g(m+sz)$ is constant in z. The target and cyclic terms give exactly

$$g(m+sz) = g(m) + Dz + 5z^2. \tag{2}$$

Let $L = 2R\alpha$ and $h = \max\{-R, \min\{R-L, -D/10 - L/2\}\}$. The interval $[h, h+L]$ is a lowest-value interval of length L for the convex quadratic. The maximizing AVaR density is $(1-\alpha)^{-1}$ on its complement and zero on that interval. The two tail moments are

$$M_1 = -\frac{L(2h+L)}{2(2R-L)}, \qquad M_2 = \frac{2R^3 - L(3h^2+3hL+L^2)}{3(2R-L)}.$$

Hence the risk equals $g(m) + DM_1 + 5M_2$. This also follows from the threshold representation of AVaR, with the threshold chosen at the boundary of the retained lower interval. Endpoint minima and equal boundary values are included by the clipping rule.

For a lower witness, any dyadic $\widehat{h} \in [-R, R-L]$ defines an exactly feasible excluded interval and a feasible tail density. Differentiating its objective holds $\widehat{h}$ fixed. This distinction is essential: a derivative of a changing approximate tail need not be the derivative of a supporting objective. Upper values use an outward enclosure of the maximizing h, while lower witnesses use the separately retained feasible $\widehat{h}$. The lower witness therefore remains valid when the two differ.

At a fixed initial state, $F(x, u, z)$ is affine in u and z. A convex continuation has a global affine minorant at every successor anchor. For a density constant on a bin, integration of this affine minorant equals its value at the midpoint. Its action dependence is affine. Adding the strong-convexity minorant of h gives (35). A fitted continuation is used only where its nonnegative convex readout makes this argument valid; no analogous assertion is made for an unrestricted nonconvex fit.

For a value upper bound, convexity places a fitted continuation below the larger endpoint value on a bin. For V_1, the midpoint upper value plus $15R/n$ bounds the whole bin by (33). If $l_i \le v(z) \le u_i$ on each bin, monotonicity of the original risk gives

$$\rho_\alpha\left(\sum_i l_i 1_{B_i}\right) \le \rho_\alpha(v) \le \rho_\alpha\left(\sum_i u_i 1_{B_i}\right).$$

The step-function risks are finite upper-tail means because the bin probabilities are exactly equal and $n(1-\alpha)$ is an integer. Their computation does not assume that v is constant on the bins.

8.3. *Complete policy and centered comparison*

PROOF OF THEOREM 11: Every fixed density is feasible for the original risk supremum. Every continuation support is below V_1 at every successor. Accordingly its all-action quadratic minimum is below V_0, irrespective of which feasible proposal supplied the anchor. The binwise upper calculation dominates the value of the

entire terminal controller; monotonicity and cash invariance charge at most $\beta\varepsilon_1$ for its terminal optimality allowance. It follows that $\underline{V}_0 \le V_0 \le J_0^{\widehat{\pi}} \le \overline{J}_0^{\widehat{\pi}}$. Subtracting proves (36).

For the centered statement take any lower and upper continuation functions ℓ, u furnished by the preceding supports and bounds. Their differences from f give admissible error functions. The exact centered identity cancels the common $\rho_\alpha(f)$ before any numerical enclosure. The preceding lower witness is a minorant of the resulting all-action expression; the preceding policy upper envelope is a majorant of the resulting policy expression. They are sufficient subsolution and supersolution inequalities in the general theorem. Computing them jointly is an implementation of that comparison, not a new objective. Loose global residual constants remain a valid alternative, but need not replace these sharper local witnesses. *Q.E.D.*

Whole-cell residual interpolation. On a state cell with vertices v_i, choose its multilinear convex weights $\omega_i(y)$. If f and V_1 are convex, their vertex upper enclosures interpolate to upper bounds. A fixed affine support L_V of V_1 satisfies

$$V_1(y) - f(y) \ge L_V(y) - \sum_i \omega_i(y)\overline{f}(v_i) = \sum_i \omega_i(y)\{L_V(v_i) - \overline{f}(v_i)\}.$$

Its right side is at least the smallest vertex difference. Maximize that lower bound over the stored supports. Reversing V_1 and f, using upper values of V_1 and lower supports of f, gives the upper residual bound. The finite maximum or minimum over anchors and cells is taken after these whole-cell inequalities. No interpolation equality for the unknown function is asserted.

Strict tail exclusion. Suppose $u_i < \tau$ and the union H of $k = n(1-\alpha)$ other bins has $v(z) \ge \tau$ throughout. It has probability at least $1-\alpha$. If a feasible maximizing density assigned positive mass to B_i, some capacity $q < (1-\alpha)^{-1}$ would remain on H. Moving equal mass from B_i to H strictly increases its risk objective, a contradiction. Equality yields only weak improvement and cannot support exclusion. Bin bounds used to construct H must come from the unpruned calculation. The argument covers a continuum of densities without enumerating all of them.

8.4. *Statistical unit and numerical allowance*

For two complete policies that share the terminal controller and differ only in their first action, the stage-action difference is at most $5/64$. Since $\sum_j b_j = 3d/8$ and the largest feasible action difference is $1/4$, (33) gives a future-value difference of at most $45/32$. Terminal-controller errors take values in $[0, \varepsilon_1]$, so their difference is at most ε_1, not twice that allowance. Monotonicity and cash invariance give the deterministic paired support

$$|J^A(x) - J^B(x)| \le B := 5/64 + \beta(45/32 + \varepsilon_1).$$

This support is established before observing a workload. It is not replaced by an observed sample range.

For n independently sampled initial states and paired cost intervals $[D_i^-, D_i^+]$, intersect each interval with $[-B, B]$. For any fixed center c computed from these intervals,

$$\frac{1}{n-1}\sum_i (D_i - \overline{D})^2 \le \frac{1}{n-1}\sum_i \max\{|D_i^- - c|, |D_i^+ - c|\}^2 =: \bar{s}^2.$$

Applying the two-sided empirical Bernstein inequality (Maurer and Pontil, 2009) to each of twelve fixed contrasts, followed by a union bound, is conservative with $\log(960) < 7$. A simultaneous sampling radius is

$$r_n = \sqrt{14\bar{s}^2/n} + \frac{98B}{3(n-1)}.$$

The resulting interval is $[n^{-1}\sum_i D_i^- - r_n, n^{-1}\sum_i D_i^+ + r_n]$, evaluated outwards. The calibration selector is fixed independently of these rows, and all pairwise contrasts are retained. Innovation bins, fitting labels, repeated sorting trials and repeated executions of an unchanged stream are not additional independent observations.

8.5. *Proof of the horizon-uniform construction*

PROOF OF PROPOSITION 12: The primitive state cost has ℓ^1 Lipschitz modulus at most $8/d$. The transition's state Jacobian has nonnegative entries with every column sum at most $11/16$. Each control exposure column has ℓ^1 norm $3d/8$. On

the outer action rectangle $[0,1/4]^m$, each component of the action-cost gradient is at most $13/16$. The scalar and vector transitions remain in the cube on that rectangle: for the vector case their range is contained in $[1/64,59/64]^d$. These bounds therefore remain applicable during the state and action comparisons even when an intermediate action is not feasible at an intermediate capacity.

Capacity intervals and capacity simplexes have Hausdorff modulus $1/(8d)$ in the action ℓ^1 norm. In the vector case, when capacity decreases, scale an action radially to the smaller capacity; the required total action displacement is no larger than the capacity change. Monotonicity and cash invariance make AVaR nonexpansive in the supremum norm. It follows that any continuation with state modulus L gives a Bellman value modulus at most

$$\frac{8}{d}+\beta\frac{11}{16}L+\frac{1}{8d}\left(\frac{13}{16}+\beta\frac{3d}{8}L\right)=\frac{1037}{128d}+\beta\frac{47}{64}L.$$

Substitution of $L=27/d$ gives $\{1037/128+705\cdot 27/1024\}/d<27/d$. The terminal cost has modulus $10/d$. This proves the uniform value modulus by backward induction. The same calculation applies to an arbitrary Lipschitz fitted continuation; convexity of that fit is not required. Consequently every own-future cone may be constructed with the common primitive modulus $27/d$.

For convexity of the original values, assume inductively that V_{t+1} is convex and its coordinate subgradients have absolute value at most $27/d$. For a smooth such function, the negative contribution from the bilinear state transition is at worst $27\|v\|_2^2/(8d)$; the Jacobian–Hessian–Jacobian term is positive semidefinite in the full state-action direction. Adding stage curvature yields

$$\frac{1}{d}\left(4-\beta\frac{27}{8}\right)\|v\|_2^2+\frac{7}{4}\|w\|_2^2=\frac{107}{128d}\|v\|_2^2+\frac{7}{4}\|w\|_2^2.$$

The action bound includes the vector cost interaction, whose off-diagonal Hessian entries equal $1/4$. For a nonsmooth convex continuation, mollify on a neighborhood of the transition image, which is a fixed positive distance from the cube boundary. Mollification preserves convexity and coordinate Lipschitz bounds there. The preceding uniform curvature inequality passes to the limit. Integration under any fixed feasible innovation density and the supremum over those densities preserve

it. Partial minimization over the convex capacity graph proves convexity of V_t. This establishes the induction without assuming differentiability of the resulting risk-averse values.

The tensor state cover has ℓ^1 covering radius at most $d/(2n_x)$. Hence the two-sided cone term is at most $27/n_x$, including the terminal cone formed from exact terminal labels. The action objective modulus is bounded by

$$D_a \le \frac{13}{16} + \beta \frac{3d}{8} \frac{27}{d} = \frac{1319}{128}.$$

The scalar feasible net has radius no larger than $A(x)/(2m)$. The vector simplex lattice has ℓ^1 radius at most $A(x)/m \le 1/(4m)$. To verify the latter, scale by $m/A(x)$, take the coordinate floors, and, when they do not exhaust the integer capacity, compare the two points obtained by rounding one coordinate up. One of their ℓ^1 errors is at most one. If the floors exhaust capacity, the point is already a lattice point. Thus the common action-net contribution is at most $1319/(512m)$.

On one innovation rectangle, the maximal ℓ^1 displacement from its midpoint under the vector transition is $d(1/(32n_z) + 1/(64n_z))$. A $27/d$-Lipschitz continuation therefore varies by at most $81/(64n_z)$. Applying the original risk's nonexpansiveness and then discounting bounds the objective integration error by $e \le 1215/(1024n_z)$. The scalar case is covered by the same bound. Exact AVaR of the midpoint step function is computed by sorting the n_z^s equally weighted values; $4 \mid n_z$ ensures integral tail counts for both risk levels. The innovation law remains continuous: the bound on e is the explicit difference between its risk and the step-function risk.

The cone proof also applies to the present risk recursion. Indeed, for any bounded u, v and scalar c, monotonicity and cash invariance give $\|\rho(u) - \rho(v)\|_\infty \le \|u - v\|_\infty$ and $\rho(u + c) = \rho(u) + c$. Feasible-action minima preserve order and this bound. Thus label, action-net and continuation-cover errors propagate by the same discounted subsolution and supersolution induction; no linear expectation or frozen maximizing density is used in that induction. The graph and action moduli established above verify its primitive hypotheses for every own-future continuation. Theorem 2, with exact feasible repair and no additional label-arithmetic error, now

bounds each nonterminal residual sum by

$$b := 27/n_x + 1319/(512m) + 1215/(512n_z).$$

The terminal cone contributes at most $27/n_x$. Thus the finite-horizon bound is $\sum_{t<T} \beta^t b + \beta^T 27/n_x \le b/(1-\beta)$, which is (38). The displayed sufficient resolutions divide the approximation allowance into three equal discounted shares. With the same uniform moduli, the sensor, repair and action-rounding coefficients in (9) are bounded independently of the horizon; their discounted sum is at most sixteen times the one-date allowance. Finite sensor and action precision can therefore fill a separately reserved positive share of the target. *Q.E.D.*

Work, storage, and exact-number growth. Every date uses S targets, at most A_m feasible actions per target and n_z^s midpoint queries per action. A dense cone query scans S labels in $O(dS)$ rational operations, giving the stated dense count. The original-owner transform costs $O(dS)$ per date; cell location and the exact off-grid owner query give the compiled count. Sorting costs $O(TSA_m n_z^s \log(n_z^s))$ comparisons and is charged in addition, not absorbed into an unmeasured primitive oracle. Stored labels and witnesses occupy $O(TSd)$ rational words, with their bit lengths charged separately.

All fixed meshes, costs, transition coefficients, risk weights and cone moduli in this construction are rational. Their finitely many primitive denominators divide a fixed integer D whose logarithm is $O(\log(dn_x m n_z))$. At a new date, a stored label is combined with fixed-denominator primitive values, cone distances, tail averages and the rational discount. A common-denominator implementation therefore increases the denominator exponent by only a fixed amount per date. Maxima, minima, comparisons and exact feasible repairs do not increase it exponentially. Bounded discounted stage costs and the fixed cone modulus also bound label magnitudes uniformly in the horizon. The resulting numerator and denominator bit lengths are $O((T+1)\log D)$. Arithmetic on these integers is not assigned constant bit cost. The statement is a fixed-accuracy constructive horizon bound with an explicit state and innovation curse, not a measured arithmetic crossover or a full implementation benchmark.

9. DECISION-SCALE QUALIFICATION: PROOFS

PROOF OF PROPOSITION 13: For every real threshold η, the function $\phi_\eta = (v - \eta)_+$ is convex and K-Lipschitz. Conditional Jensen and the chord inequality give $M_n\phi_\eta \le P\phi_\eta \le E_n\phi_\eta$. Apply the AVaR threshold formula $\inf_\eta\{\eta + (1-\alpha)^{-1}\mu\phi_\eta\}$ for each of the three probability measures. Taking infima preserves the two orders and proves the enclosure. In particular, no interchange of an action infimum and a risk supremum occurs.

Here is a uniform bound which remains valid at kinks. Let $h = 2R/n$. Choose one-sided slopes of ϕ_η at each bin's outer endpoints. The slope increases across each bin; its total increase over the whole interval is at most $2K$. The difference between the endpoint average and midpoint value on bin i is no greater than $h/4$ times the slope increase across that bin. This follows by integrating the left and right slopes over the two half-bins. Choose the left endpoint right derivative and right endpoint left derivative; jumps at shared endpoints may be omitted, which only decreases the sum of within-bin slope increases. Therefore

$$0 \le E_n\phi_\eta - M_n\phi_\eta \le \frac{h}{4n}(2K) = \frac{KR}{n^2}.$$

For completeness, the half-bin expression equals one half the integral of the difference between the right and left slopes at equal distances from the midpoint, so its upper bound also follows directly without assuming a classical second derivative. The preceding uniform bound passes through the two infima and gives (39). Monotonicity and cash invariance add $e_- + e_+$ for endpoint error. They also charge the actual terminal controller's one-sided error only on the upper side.

For a product law, apply the one-coordinate midpoint–endpoint comparison successively, holding the other coordinates fixed. Products of positive averaging operators preserve inequalities and telescope. Each partially averaged positive-part function remains convex and has the same Lipschitz modulus in each unintegrated coordinate. Summing the one-coordinate error bounds and then taking the threshold infimum proves the product statement. This argument uses neither a discrete economic transition assumption nor differentiability of v. *Q.E.D.*

PROOF OF THEOREM 15: Every service's regret belongs to $[0,\tau]$ at every possible initial state. The difference of two complete-policy values therefore belongs to $[-\tau,\tau]$, because the common original optimal value cancels. Each interval intersection retains the true paired difference. A difference of two intervals of width at most w has width at most $2w$. Intersecting with a valid support cannot increase that width.

Hoeffding's inequality for a variable of range length 2τ gives

$$\Pr\{|\overline{D} - \mathbb{E}D| > r\} \leq 2\exp\{-Nr^2/(2\tau^2)\}.$$

A union bound across the H fixed contrasts gives the stated coverage. Pairwise contrasts need not be mutually independent; only the initial states within each contrast must be independent, conditional on the frozen preparation. Replacing the unobserved sample mean by its numerical lower and upper bounds yields (40). Their difference is at most $2(r+w)$. This is an application of the established concentration inequality (Hoeffding, 1963), not a claim to a new concentration principle.

For the chosen rational parameters, $Nr^2/(2\tau^2) = 8$. The inequality $24e^{-8} < 1/20$ can be checked entirely with rational arithmetic: the partial sum $\sum_{j=0}^{11} 8^j/j!$ is greater than 480. Equation (41) is an exact rational comparison. In numerical classification the margin is rounded inward and the reported confidence endpoints outward. No observed range or variance is used to choose the radius. *Q.E.D.*

Totalization of the admitted service. The first-date objective with the original V_1 has action Lipschitz constant less than 7: the primitive action derivative is at most $3/4$, and the discounted continuation contribution is at most $(15/16)(15/d)(3d/8)$. The feasible interval has length at most $1/4$. A uniform net with $m = 1024$ intervals therefore gives an all-action lower value by subtracting $7/(8m)$ from the smallest grid cost lower bound. The grid point with the smallest upper value supplies a feasible policy upper value.

The exact fallback uses $n = 256$ innovation bins and the preceding convex-order rule. Exact one-date rational supports have gaps at most 2^{-30}; the terminal controller contributes 2^{-20}. The resulting quadrature width, including these allowances, is

below $1/8192$ for both risk levels. Adding $7/(8m)$ leaves the original-optimum bound below $1/256$. Downward rounding of the first action to 40 bits costs at most 72^{-40} on each side and fits strictly inside both remaining allowances. All values, risk weights, feasible actions and endpoint computations in this final construction are rational, and the terminal strong-convexity bisection reaches its stated gap in finitely many steps. Thus the fallback defines a finite valid action at every dyadic initial observation in the target distribution. For arbitrary real initial states, the separate acquisition transfer remains required; the confirmation theorem here concerns the declared dyadic initial law.

The floating implementation may reach the two targets much sooner. That does not permit removing the exact fallback from the policy definition, treating rejected proposals as missing observations, or asserting that the fallback was exercised when the actual counter is zero. Hardware or resource failure aborts the experiment and is recorded as a failure, not a change in its conditional sampling law.

PROOF OF PROPOSITION 16: Subtract the two complete acquisition-and-deployment accounts. Using the upper policy, acquisition and fee bounds and the lower computational saving bounds above (42) by $m(d^+ - \lambda s^-) + \lambda k^+ + F^+$. Nonpositivity gives (43). Solving this scalar inequality proves the integer threshold and its sign qualifications. The separate quality inequality depends on policy cost rather than price and cannot be deduced by increasing λ. A common audit cost cancels only when it is actually common and purchased in both alternatives; otherwise its incremental charge remains in k^+ or F^+. *Q.E.D.*

PROOF OF COROLLARY 14: Write $h = 2R/n$ and $k = n(1-\alpha)$. The two roots of $v_n - \eta_n$, where $\eta_n = K\alpha R$, are $c_n - \alpha R$ and $c_n + \alpha R$. They lie strictly inside the domain and at two distinct bin midpoints, since αn is an even integer. Under M_n the mass strictly above η_n is $(k-1)/n$ and the mass at η_n is $2/n$. Under E_n there is no mass at η_n, and the mass above it is exactly k/n. Hence η_n minimizes the AVaR threshold objective under both measures, by its left and right slope conditions.

The hinge $\phi = (v_n - \eta_n)_+$ is affine on every bin except the two whose midpoints are its roots. On either exceptional bin, the endpoint average minus the midpoint value is $Kh/4$. Each bin has weight $1/n$. The two threshold minima therefore differ

by

$$(1-\alpha)^{-1}(E_n\phi - M_n\phi) = \frac{1}{1-\alpha}\frac{2}{n}\frac{Kh}{4} = \frac{KR}{(1-\alpha)n^2}.$$

All equalities use the endpoint mixture's half weights at the outer boundaries. This explicit sequence establishes sharpness of the stated enclosure on its function class; it does not assert that the investment value itself equals v_n. *Q.E.D.*

10. INFORMATION BOUNDS AND CERTIFICATE-ORIENTED ACQUISITION

10.1. *Proof of Theorem 17*

Put $q = -R + 2R\alpha$ and $L = R - q = 2R(1-\alpha)$. Answer the algorithm's queries with

$$f_0(z) = \frac{K}{2L}(z-q)_+^2.$$

This function is convex and nondecreasing. Its derivative is zero to the left of q, and equals $K(z-q)/L$ to the right, so its Lipschitz constant on $[-R,R]$ is K. After the algorithm terminates, collect all distinct query points in (q,R) and augment them by q and R. They partition the tail interval into $s \le N+1$ nonempty intervals of lengths $h_1,\dots,h_s$.

Let $f_1 = f_0$ on $[-R,q]$ and let f_1 equal the chord of f_0 on each interval of this partition. For endpoints $a < b$ in $[q,R]$, the chord slope is $K(a+b-2q)/(2L)$. These slopes increase along the partition, lie in $[0,K]$, and are compatible with the zero left derivative at q. Hence $f_1 \in \mathcal{C}_{K,R}$ and is nondecreasing. The two functions agree at every query point, including any points outside the tail. By induction on the number of queries, a deterministic adaptive algorithm receives exactly the same transcript on f_0 and f_1, chooses the same next points, and stops with the same output. The partition is constructed after running the algorithm on f_0; the argument therefore does not assume nonadaptive sampling.

For any nondecreasing continuous f, the quantile representation of AVaR gives

$$\mathrm{AVaR}_\alpha(f(Z)) = \frac{1}{L}\int_q^R f(z)\,dz.$$

This also holds at $\alpha = 0$ and in the presence of the flat portion of these functions. On an interval $[a, b]$ of length h, the chord minus the quadratic equals $K(z-a)(b-z)/(2L)$. Its integral is $Kh^3/(12L)$. Consequently,

$$\begin{aligned}\mathrm{AVaR}_\alpha(f_1(Z)) - \mathrm{AVaR}_\alpha(f_0(Z)) &= \frac{K}{12L^2}\sum_{i=1}^{s} h_i^3 \\ &\geq \frac{K}{12L^2}\frac{(\sum_{i=1}^{s} h_i)^3}{s^2} \geq \frac{KL}{12(N+1)^2} = \frac{KR(1-\alpha)}{6(N+1)^2}.\end{aligned}$$

The first inequality follows from convexity of the cube. A valid interval returned on the common transcript must contain both values. A point estimate has error at least half their separation on one input. The midpoint–endpoint upper bound already proved in the article uses n midpoint values and $n+1$ distinct endpoint values. Taking $n = \lfloor (N-1)/2 \rfloor$ for $N \geq 3$ gives the stated upper order. The constants on the two sides have different α dependence; the conclusion concerns the exponent at fixed α, not uniform constant optimality. *Q.E.D.*

The construction deliberately permits exact function values to the lower-bound algorithm. Thus any finite-precision procedure with no additional information cannot evade the bound by using more accurate arithmetic. Conversely, this lower bound does not price arithmetic, optimization, or the work of acquiring a continuation formula; those costs enter the service ledger. Providing the entire piecewise representation of f for free would change the information model and invalidate this comparison.

10.2. *Proof of Proposition 18*

For $g(x) = a(x_1) + b(x_2)$ the corner difference

$$g(1,1) - g(1,0) - g(0,1) + g(0,0)$$

is zero. For v it equals one. If $\|v - g\|_\infty \leq e$, the triangle inequality gives $1 \leq 4e$. Equality is attained by $g(x) = (x_1 + x_2)/2 - 1/4$, because with $s = x_1 + x_2 \in [0,2]$ the error is $1/4 - s/2$ for $s \leq 1$, and $s/2 - 3/4$ for $s \geq 1$.

For a total-degree-two polynomial, restrict to $x_1 = x_2 = (u+1)/2$, $u \in [-1,1]$. If p approximates u_+ with error at most e, then $r(u) = 2p(u) - u$ approximates $|u|$ with error at most $2e$. Symmetrizing r does not increase the error and produces an even polynomial $au^2 + c$. The linear functional

$$\mathcal{L}f = f(1/2) - \tfrac{1}{4}f(1) - \tfrac{3}{4}f(0)$$

annihilates every such polynomial, has sum of absolute weights two, and satisfies $\mathcal{L}|u| = 1/4$. Therefore its approximation error for $|u|$ is at least $1/8$, and $e \geq 1/16$. The polynomial

$$p(x) = \tfrac{1}{2}\{u + u^2 + 1/8\}, \qquad u = x_1 + x_2 - 1,$$

attains the bound: $|u| - u^2$ ranges from zero to $1/4$. Finally, v is precisely one ReLU with weight $(1,1)$ and offset -1. *Q.E.D.*

The distinction between function and decision error matters. Cash shifts do not affect an action choice, and even a nonconstant approximation error may preserve the minimizer. No lower bound for policy regret follows from this proposition without a further decision construction. In particular, the proposition is not used to reinterpret the R96 quadratic winner as a neural winner.

10.3. *Proof of Proposition 19*

By definition, $L(h) \leq V(z) - h(z) \leq U(h)$ on the entire covered domain. Substitute $z = F(\omega)$, use monotonicity of ρ, and then use cash invariance to obtain (46). After the numerical candidate is rounded, the whole-domain enclosure is recomputed for those *actual* coefficients. The selection rule compares two valid enclosure widths and chooses the smaller one. This proves the width statement independently of the linear-program solver's termination tolerance. *Q.E.D.*

For completeness, a general finite-site extension makes the distinction between node and domain certificates explicit. Suppose V and h have known Lipschitz constants L_V and L_h, the sites have covering radius δ, and $V(z_i) \in [\ell_i, u_i]$. Then

$$\inf_z (V - h)(z) \geq \min_i \{\ell_i - h(z_i)\} - (L_V + L_h)\delta,$$

$$\sup_z (V-h)(z) \le \max_i \{u_i - h(z_i)\} + (L_V + L_h)\delta.$$

Indeed, choose a nearest site and apply the Lipschitz inequality to $V-h$. Directed rounding must enclose the displayed arithmetic. The implementation retains the stronger inherited convex-cell residual construction rather than replacing it by this generic bound. Its linear program is only a candidate generator; both the baseline and candidate are evaluated by the same whole-domain validator.

The LP fixes the intercept to zero, bounds the two affine slopes in $[-10^4, 10^4]$, and bounds nonlinear coefficients in $[0, 10^4]$. These are explicit acquisition restrictions, not claimed characterizations of the best neural approximation. The two residual endpoints are unrestricted. Setting all readout coefficients to zero and choosing endpoints to enclose all acquired value intervals is feasible, so a solver failure is an execution failure, not evidence that the mathematical program is infeasible. The retained nonnegative nonlinear coefficients remain convex after dyadic rounding. All attempted feature fits, rank rejections, selection ties, and LP iterations are retained in construction receipts.

10.4. *Proof of Theorem 20*

Subtract total candidate cost $m\{B\mathbb{E}J^a + \lambda c_a\} + \lambda K_a + F$ from comparator cost $m\{B\mathbb{E}J^b + \lambda c_b\} + \lambda K_b$. This gives (47). Monotonicity of this affine expression in each interval argument, using $m, B \ge 0$ and $\lambda \ge 0$, gives (48)–(49). The strict inequality $mq > C$ is equivalent to $m > C/q$ when $q > 0$, yielding the first integer stated in the theorem. *Q.E.D.*

If $q \le 0$ and $C \ge 0$, this sufficient lower-bound test never certifies adoption at a positive workload count. That is an unresolved or negative *comparison*, not a failure of the research program. An upper bound strictly below zero certifies rejection for the specified prices and workload. An equality at zero is kept distinct from either strict conclusion. When each service satisfies $0 \le J^a(x) - J^*(x) \le \tau$, subtraction gives $-\tau \le J^a(x) - J^b(x) \le \tau$ pointwise and hence after expectation. No independence assumption is needed for this deterministic bound. For sample-based refinements, independence concerns initial-state draws; the paired services

intentionally share each initial state and their deterministic original-law integration. The factor B cannot be dropped merely because the services share these draws.

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